Beyond Closed-Set Accuracy: Constituent-Level Generalization in Compound Motor Fault Diagnosis

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***Abstract*—Reported accuracies for data-driven motor fault diagnosis are frequently near ceiling, yet they rarely measure the ability that industrial deployment actually requires: recognizing the elementary fault mechanisms of a compound fault whose exact combination was never observed during model development. This paper evaluates that ability on a public multimodal dataset of a 2.2-kW three-phase induction motor containing 24 diagnostic states, including nine compound faults that pair a mechanical bearing defect with a second mechanical, electrical, or electromagnetic mechanism, under twelve speed/load conditions. The problem is formulated at the constituent level: nine elementary mechanisms are detected independently from physics-guided, mechanism-specific vibration- and current-domain features, so unseen combinations remain expressible. A crossed protocol separates three levels of compound-context exposure (recombination, one-sided, and two-sided context zero-shot) from operating-condition novelty, with run-level, severity-aware splits and all calibration confined to training data. Although closed-set 24-class accuracy reaches 0.913, exact recovery of unseen constituent sets remains between 0.083 and 0.157 for the locked linear probe—not clearly above the 0.111 achieved by the strongest input-independent constant predictor—and the constituent metrics are reported against such prevalence references throughout, which expose several partial-recovery scores as prevalence-driven; the probe's genuine advantage is a joint operating point of higher micro-F_1 (0.631–0.686 versus a best-constant 0.556) at comparable or lower false-addition cost. Compound-context deprivation is measurably harmful chiefly when the operating condition is also unseen (recombination versus two-sided context zero-shot: micro-F_1 interaction $+0.040$, 95% CI $[0.009, 0.073]$), and cross-partner transfer is strongly directional, including one statistically supported score-orientation reversal whose physical attribution is deliberately left open given demonstrated recording-session separability. Threshold-objective and classifier-family analyses show that exact decomposition, not constituent recall, is the binding constraint.**

***Index Terms*—compound faults, condition monitoring, constituent-level diagnosis, data leakage, electrical machines, evaluation protocols, fault diagnosis, induction motors, multi-label classification, zero-shot learning**

I. Introduction

INDUCTION motors underpin industrial drivetrains, and their unplanned failures propagate directly into downtime and safety risk. Data-driven condition monitoring has consequently produced a large literature whose reported accuracies now cluster near ceiling on standard laboratory benchmarks. Deployment experience is harder to reconcile with those numbers: a model commissioned at one operating point degrades at another, and faults in service rarely arrive one at a time. A bearing defect that develops alongside a winding short circuit, broken rotor bars, or eccentricity confronts the monitoring system with a *compound* state whose exact combination was, in general, never present in the training history.

Conventional closed-set formulations cannot even express this situation: a previously unseen combination $A+B$ constitutes an unseen class. The compound-fault literature therefore moved toward decoupling formulations that predict elementary mechanisms rather than combinations, and, more recently, toward zero-shot settings in which compound states must be recognized from single-fault training data alone. As Section II reviews, however, that literature has evaluated almost exclusively *homogeneous* compounds—bearing–bearing or bearing–gear pairings whose constituents are all observable in the same vibration channel—and its reported accuracies of roughly 75–87% have been obtained under evaluation practices that a parallel line of work on data leakage has shown to be fragile: sample-level splits, steady operating conditions, and metrics reported without reference to what label prevalence alone would achieve.

This paper asks a deliberately harder question on a public dataset that makes it answerable. The MCC5-THU motor dataset [1] provides 288 recordings of a 2.2-kW three-phase induction motor across 24 diagnostic states under twelve speed/load conditions, with synchronized triaxial vibration, three-phase currents, torque, and a key-phase tachometer. Its nine compound faults pair a mechanical bearing defect with a second mechanism—inner or outer race, broken rotor bars, static or dynamic eccentricity, or a stator winding short circuit—so that, in eight of the nine combinations, the second constituent is electromagnetic in origin, and for the winding and rotor-bar constituents its primary evidence resides in the stator currents rather than in vibration. To our knowledge, neither this cross-modal compound regime nor this dataset has previously received a constituent-level evaluation.

The study is designed so that every claim is bounded by an explicit control. The formulation is constituent-level: nine elementary mechanisms are detected independently by regularized linear probes operating on curated, mechanism-specific physical-signature families derived from angle-resampled order spectra, envelope-order descriptors, and phase-current sequence measures. A crossed protocol varies compound-context

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exposure over three regimes—recombination, one-sided, and two-sided compound-context zero-shot—and crosses each with seen and unseen operating conditions, under run-level, severity-aware splits with all preprocessing, calibration, and threshold selection confined to training data. Every reported metric is accompanied by input-independent constant-predictor references, so that prevalence-driven scores cannot masquerade as diagnostic skill; oracle detectability analyses are paired with recording-sensitivity and same-date controls, so that separability is not silently attributed to fault mechanisms; and paired run-clustered bootstrap inference with multiplicity correction distinguishes supported effects from noise.

The contributions are as follows.

1) **A constituent-level formulation and crossed composition–context–condition protocol** for compound motor faults whose constituents span mechanical and electrical modalities, separating novel composition, compound-context deprivation, and operating-condition novelty in one design (Sections III–IV).
2) **Prior-aware evaluation**: all constituent metrics are reported against input-independent prevalence references, which expose several apparently favorable partial-recovery metrics as at or below what a constant predictor achieves. Against the best constant reference per metric, the locked probe's advantage is a joint operating point—higher micro-F_1 at comparable or lower false-addition cost, with all-constituent recovery exceeding its floor under recombination—rather than uniform per-metric superiority.
3) **A controlled analysis of constituent transfer**: within-pair oracle detectability, cross-partner transfer, physical-signature preservation, and a statistically supported score-orientation reversal, each bounded by recording-sensitivity and same-date controls that keep session effects from being read as physics.
4) **Sensitivity and robustness analyses** over the decision-threshold objective, single-fault severity merging, and four classifier families, showing that the qualitative generalization pattern is stable while absolute exact-match decomposition is an operating-point and model property, not an intrinsic bound of the protocol.
5) **Documentation of dataset-release properties that materially affect protocol design**—recording-session separability and severity-variant leakage paths—together with reference operating-condition experiments (Appendix A) that motivate the crossed design.

The principal findings are that closed-set discriminability (accuracy 0.913) coexists with exact constituent-set recovery of 0.083–0.157 on unseen compounds for the locked probe; that recovery of at least one constituent is nearly universal while exact decomposition fails, so the practical failure mode is incomplete decomposition with over-diagnosis rather than silence; that compound-context deprivation becomes measurably harmful specifically under simultaneous operating-condition novelty; and that constituent evidence is often, but directionally and heterogeneously, transferable across compound partners.

The remainder of the paper is organized as follows. Section II reviews related work and positions the study. Section III details the constituent representation, feature construction, detection, protocol, metrics, and statistical analysis. Section IV reports results. Section V concludes. Appendix A reports the operating-condition reference experiments.

II. Related Work

Several research threads intersect in this study: compound-fault decoupling, zero-shot recognition of unseen fault combinations, the reliability of evaluation protocols in data-driven condition monitoring, generalization across operating conditions, multimodal motor diagnosis, and the benchmark datasets that underpin them. We review each and then position the present work (Table I).

Early data-driven treatments of compound faults assigned each observed combination its own class, which cannot express combinations absent from training. Decoupling formulations address this by predicting the elementary mechanisms instead. Huang *et al.* proposed a deep decoupling convolutional network whose multi-label classifier recovers the constituents of gearbox compound faults from raw vibration signals [2], and a one-dimensional CNN variant with a multi-label output layer [3]. Subsequent work refined the decoupling head: capsule networks with dynamic routing and threshold-based label activation [4], attentional residual networks combined with active learning for measured (rather than injected) bearing compound faults [5], and mixture-of-experts architectures with task-adapted attention for decoupling compounds unseen as combinations [6]. Two properties are near-universal in this line: the constituents are *mechanical* (bearing races, balls, gear teeth), so every constituent is observable in the same vibration channel; and evaluation is sample-level on steady-speed laboratory recordings, usually with segments of the same recording on both sides of the train/test split.

A stricter setting withholds *all* compound data and requires composing single-fault knowledge at test time. Xu *et al.* introduced zero-shot compound-fault diagnosis with manually defined semantic attributes learned from single-fault data [7], later replacing hand-crafted attributes with autoencoder-learned fault semantics [8]. Generative formulations synthesize pseudo-samples of unseen compounds from single-fault semantics [9], [10]; graph embeddings propagate relational structure among base faults [11]; and physics-informed semantic spaces anchor the attribute vectors in characteristic-frequency knowledge [12]. Reported accuracies on unseen compounds cluster between roughly 75% and 87% [7], [8], [12], [13].

These results, however, concern compounds whose constituents are *homogeneous*: bearing–bearing or bearing–gear combinations in which both mechanisms modulate the same vibration carrier. The compounds in the MCC5-THU motor dataset instead pair a mechanical bearing defect with an electrical or electromagnetic fault (winding short circuit, broken rotor bars, eccentricity), whose primary evidence lies in the stator currents rather than in vibration. To our knowledge, this *cross-modal* compound regime has not been evaluated in the zero-shot compound-fault literature, and no published constituent-level results exist for this dataset.

A parallel literature shows that evaluation-protocol choices, more than model choices, drive many reported near-ceiling accuracies. Wheat *et al.* quantified the impact of intra-bearing data leakage in vibration diagnosis and advocated part-to-part splits [14]; recent work formalizes the distinction between segmentation-level and specimen-level leakage and proposes leakage-safe protocols [15], [16]; and open benchmark studies made similar observations for deep architectures across public datasets [17]. This thread has concentrated on single-fault classification. Its intersection with compound-fault evaluation is, to our knowledge, unexplored—yet it is precisely where leakage is most subtle: in the dataset used here, severity variants of one physical combination can leak across a composition boundary, and compound recordings are acquired in different sessions from their single-fault references, which confounds naive "compound versus single" contrasts. Our protocol design (run-level, severity-aware splits; session-matched sensitivity controls; constituent-prevalence floors for every reported metric) extends leakage-aware evaluation to the compositional setting.

Diagnosis under variable speed and load has produced a large domain-adaptation and domain-generalization literature, surveyed in [18]. On the MCC5-THU family specifically, the dataset authors proposed evidential ensemble learning for real-time multimode diagnosis [19] and recently questioned how operating conditions should enter learning-based diagnosis at all [20]; other groups applied condition-robust architectures to the companion gearbox release [21], [22]. The dominant protocol in this literature holds one condition out of many [18], [23]. Our reference experiments on this dataset (Appendix A) show that protocol to be nearly saturated—training on eleven of twelve conditions costs only $\sim$3 accuracy points, because the remaining conditions bracket the held-out one, whereas the operationally realistic reverse direction (training on a single condition) costs over 60—which motivates the crossed composition–context–condition design used here rather than leave-one-condition-out alone.

That mechanical faults imprint on stator currents and electrical faults on vibration has long motivated multimodal motor diagnosis [24]. Deep multimodal fusion typically concatenates modalities into one representation or fuses learned features, e.g. multilevel information fusion [25], dynamic-routing multimodal networks [26], adaptive CNN fusion of denoised vibration and current [27], and comparative studies of the two modalities under deep and classical learners [28]. These works report that fusion improves *single-fault* accuracy. The compound setting reverses the picture: the reference experiments of Appendix A show that a shared feature space lets the dominant (mechanical) signature suppress the electrical one—a winding short detectable from currents alone on 39.3% of compound windows falls to 0.2% once vibration features are concatenated. The mechanism-specific feature families used here are a structural answer: each constituent detector sees only the physically appropriate modality, so cross-modal suppression cannot occur inside a detector.

Public rotating-machinery datasets underpin this literature: CWRU [29] and Paderborn [30] for bearings, and more recent multi-fault motor and drivetrain releases such as a PMSM stator-fault dataset [31] and an industrial-scale motor–pump dataset [32]. The MCC5-THU family—a gearbox release [33] and the motor release used here [1]—is distinctive in combining single *and* compound faults, twelve speed/load profiles with transitional segments, and synchronized vibration, three-phase current, torque, and key-phase channels. The motor release is distributed without reference baselines; the present study provides, to our knowledge, its first systematic constituent-level evaluation, together with documentation of release-level inconsistencies that materially affect protocol design.

Table I contrasts this study with representative prior work along the axes developed above. The distinguishing combination is: compound faults whose constituents live in *different* modalities; composition, compound-context, and operating-condition generalization varied *jointly* in one crossed protocol; run-level, severity-aware, leakage-safe splits; metrics reported against constituent-prevalence floors so that prior-driven scores cannot masquerade as diagnostic skill; and run-clustered bootstrap inference with multiplicity control.

III. Materials and Methods

The proposed study was designed to evaluate whether the elementary fault mechanisms forming a compound motor fault can be recovered when their exact combination has not been observed during model development. Rather than treating each compound condition as an independent multiclass label, the problem was formulated at the constituent-mechanism level. The methodology consisted of a primary crossed constituent-generalization experiment together with complementary analyses designed to distinguish diagnostic failure from absence, suppression, or context-dependent reorientation of constituent evidence. The analysis pipeline included constituent-level fault representation, physics-guided multimodal feature construction, independent constituent detection, crossed evaluation of composition, compound-context, and operating-condition generalization, within-pair and cross-partner transferability analysis, physical-signature preservation analysis, and paired statistical evaluation. All preprocessing, feature normalization, classifier fitting, and decision-threshold calibration were performed exclusively within the corresponding training data to prevent information leakage.

A. Constituent-Level Fault Representation and Problem Formulation

Let the dataset contain N independent experimental runs. Instead of assigning each run to one of the original fault-condition classes, every run was represented by a binary constituent vector

$$\mathbf{y}_i = [y_{i,1}, y_{i,2}, \ldots, y_{i,K}]^T, \qquad \mathbf{y}_i \in \{0,1\}^K, \qquad (1)$$

TABLE I
POSITIONING OF THIS STUDY AGAINST REPRESENTATIVE PRIOR WORK. ✓ = EXPLICITLY INCLUDED; – = ABSENT, NOT APPLICABLE, OR NOT REPORTED. "CROSS-MODAL" MEANS THE COMPOUND'S CONSTITUENTS ARE OBSERVABLE IN DIFFERENT SENSING MODALITIES (MECHANICAL + ELECTRICAL). "PRIOR-AWARE FLOORS" MEANS METRICS ARE REPORTED AGAINST LABEL-PREVALENCE (TRIVIAL CONSTANT-PREDICTOR) REFERENCES.

Study	Task focus	Signals	Compound faults	Cross-modal	Unseen combin.	Condition shift	Leakage-safe run-level splits	Prior-aware floors
Huang *et al.* [2]	multi-label decoupling	vib.	✓	–	–	–	–	–
LTSS-BoW-CapsNet [4]	capsule decoupling	vib.	✓	–	–	–	–	–
HeMTAN [6]	unseen-compound decoupling	vib.	✓	–	✓	–	–	–
Xu *et al.* [7]	zero-shot compounds	vib.	✓	–	✓	–	–	–
Xu *et al.* [8]	learned fault semantics	vib.	✓	–	✓	–	–	–
Song *et al.* [12]	GZSL, physics semantics	vib.	✓	–	✓	–	–	–
Wheat *et al.* [14]	leakage in evaluation	vib.	–	–	–	–	✓	–
Han *et al.* [18]	multi-condition survey	multi	–	–	–	✓	–	–
Liu *et al.* [19]	multimode diagnosis	multi	–	–	–	✓	–	–
DRMNN [26]	multimodal fusion	vib.+cur.	–	–	–	–	–	–
This work	constituent-level generalization	vib.+cur. (+key-phase)	✓	✓	✓	✓	✓	✓

where i denotes the experimental run and $K = 9$ denotes the number of elementary fault mechanisms considered in this study. The constituent set was

$$\mathcal{F} = \{f_{\text{ball}}, f_{\text{inner}}, f_{\text{outer}}, f_{\text{bend}}, f_{\text{bar}}, f_{\text{dyn}}, f_{\text{stat}}, f_{\text{vu}}, f_{\text{wind}}\}, \tag{2}$$

corresponding to bearing-ball, bearing-inner-race, bearing-outer-race, shaft-bending, broken-bar, dynamic-eccentricity, static-eccentricity, voltage-unbalance, and winding-related fault mechanisms, respectively.

For a healthy run,

$$\|\mathbf{y}_i\|_0 = 0, \tag{3}$$

whereas a single-fault run satisfies

$$\|\mathbf{y}_i\|_0 = 1. \tag{4}$$

A compound fault is represented by the simultaneous activation of two or more constituents,

$$\|\mathbf{y}_i\|_0 \geq 2. \tag{5}$$

The composition associated with the ith run was therefore defined as

$$\mathcal{C}_i = \{f_k \in \mathcal{F} : y_{i,k} = 1\}. \tag{6}$$

For example, a simultaneous inner-bearing and winding fault was encoded as

$$y_{i,\text{inner}} = 1, \qquad y_{i,\text{wind}} = 1, \tag{7}$$

while all remaining constituent entries were zero.

This formulation differs fundamentally from conventional closed-set multiclass diagnosis. In a multiclass formulation, a previously unseen compound such as $A{+}B$ constitutes an unseen class. In the proposed constituent formulation, the diagnostic objective is instead to recover the known elementary mechanisms A and B independently. Consequently, the primary research question becomes whether constituent knowledge learned from single faults and other compound contexts can be transferred to a novel composition.

B. Physics-Guided Multimodal Feature Representation

The diagnostic representation was derived from three vibration channels and three phase-current channels sampled at 12,800 Hz. Each 2-s analysis window was angle resampled using the measured keyphase pulses at 256 samples per shaft revolution before order-domain feature extraction. Window-level descriptors were subsequently aggregated to one run-level representation using the median across valid windows. Interquartile-range descriptors were also generated during feature extraction, but the curated mechanism-specific signatures used in the final constituent detectors excluded them. Thus, all classifiers operated on one physics-guided feature vector per experimental run. The resulting multimodal representation was written as

$$\mathbf{x}_i = \left[\left(\mathbf{x}_i^{(v)}\right)^T, \left(\mathbf{x}_i^{(c)}\right)^T\right]^T, \tag{8}$$

where $\mathbf{x}_i^{(v)}$ and $\mathbf{x}_i^{(c)}$ denote vibration- and current-derived descriptors, respectively.

For an angle-domain signal $x[n]$, the order spectrum $P_x(o)$ was estimated using a Hann-window periodogram. Around a target shaft order o_0, local spectral energy was accumulated within a ± 0.12 order band and normalized by the total spectral energy between 0.25 and 60 shaft orders,

$$R_x(o_0) = \frac{\sum\limits_{|o-o_0|\leq 0.12} P_x(o)}{\sum\limits_{0.25\leq o\leq 60} P_x(o) + \epsilon}. \tag{9}$$

Both the local band energy and its logarithmic normalized ratio, $\log_{10}[\max(R_x(o_0), \epsilon)]$, were retained in the feature bank.

For the SKF 6205 bearing, the characteristic shaft orders were

$$o_{\text{BPFO}} = 3.585, \qquad o_{\text{BPFI}} = 5.415, \qquad o_{\text{BSF}} = 2.357, \tag{10}$$

together with their second harmonics. Bearing-fault descriptors were extracted from the Hilbert-envelope vibration order spectrum.

Current-domain descriptors were obtained from the individual phase-current order spectra and the magnitude of the Clarke space vector. The dominant electrical order was identified from the maximum summed three-phase spectral energy over 0.5–30 orders. Relative to this data-derived carrier o_e, features were evaluated at

$$o_e, \qquad o_e \pm 1, \qquad o_e \pm 2, \qquad 2o_e. \tag{11}$$

The $o_e \pm 1$ and $o_e \pm 2$ components were used as electrical-carrier sideband proxies and were not interpreted as a fully slip-parameterized broken-bar model.

Current imbalance was characterized using the coefficient of variation of the three phase RMS values and a phase-order-invariant sequence ratio. Let S_1 and S_2 denote the two sequence candidates obtained from the three complex phase-current coefficients at o_e. Because the dataset channel phase order was not assumed *a priori*, the implemented sequence descriptor was

$$r_{\mathrm{seq}} = \frac{\min\left(|S_1|, |S_2|\right)}{\max\left(|S_1|, |S_2|\right) + \epsilon}. \tag{12}$$

Accordingly, this descriptor represents a phase-order-invariant sequence imbalance rather than the magnitude of a prespecified negative-sequence phasor.

Rather than exposing every constituent detector to the complete feature bank, a fixed mechanism-specific mapping was used. Bearing-inner, bearing-outer, and bearing-ball detectors used BPFI/2BPFI, BPFO/2BPFO, and BSF/2BSF envelope-order descriptors, respectively. Shaft bending used raw shaft-$1\times$ and shaft-$2\times$ vibration features. Broken-bar detection used electrical-carrier sideband proxies; dynamic and static eccentricity used combined shaft-order and carrier-sideband descriptors; voltage unbalance used sequence-ratio and phase-RMS imbalance features; and winding faults used sequence-ratio, phase-RMS imbalance, and electrical second-harmonic descriptors. For constituent k,

$$\mathbf{x}_i^{(k)} = \mathbf{M}_k \mathbf{x}_i, \tag{13}$$

where $\mathbf{M}_k$ denotes the predefined mechanism-specific feature selector. The selected sets contained 16 descriptors for each bearing mechanism and shaft bending, 40 for broken bar, 56 for each eccentricity mechanism, 2 for voltage unbalance, and 12 for winding. The unequal selector sizes reflect the number of physically distinct channel–order combinations represented by each predefined signature family rather than post-hoc optimization of feature-set dimensionality. No outer-test performance was used to modify the feature definitions or selector sizes.

C. Independent Constituent Detection

A separate binary detector was trained for each constituent mechanism, allowing multiple fault mechanisms to be activated simultaneously without a softmax constraint. For the locked primary analysis, each detector used L_2-regularized logistic regression with $C = 1$ and class balancing applied within the corresponding training data. The predefined mechanism-specific feature subset $\mathbf{x}_i^{(k)}$ described in Section III-B was used independently for constituent k.

All preprocessing was fitted strictly within each outer-training fold. Missing values were replaced using training-set medians, after which features were standardized using training-set statistics. The resulting probability $p_{i,k}$ was converted to a binary decision according to

$$\hat{y}_{i,k} = \mathbb{I}\left(p_{i,k} \geq \tau_k\right), \tag{14}$$

where τ_k is a constituent-specific threshold.

Thresholds were calibrated exclusively within the outer-training data using grouped inner cross-validation over operating conditions. Candidate thresholds from 0.05 to 0.95 were evaluated using constituent-level F_1 as the primary objective, with balanced accuracy and proximity to 0.5 used as deterministic tie breakers. The selected threshold was then fixed for the corresponding outer-test fold. No outer-test sample was used for imputation, normalization, classifier fitting, or threshold selection.

A deliberately simple linear classifier was retained as the locked primary diagnostic probe because the objective was to characterize constituent transferability rather than to optimize model architecture. This choice limits architecture-induced confounding when comparing compound-context regimes. Alternative classifier families were evaluated separately as post-hoc robustness analyses.

D. Severity-Merging Sensitivity Analysis

The locked mechanism-level formulation merges high- and low-severity single-fault examples belonging to the same constituent. Because the released compound states predominantly contain the high-severity variants of the severity-coded mechanisms, a post-hoc sensitivity analysis was performed to determine whether low-severity single-fault support materially affected the crossed generalization results. The analysis removed the low-severity single-fault runs for bearing inner, bearing outer, static eccentricity, and winding, corresponding to 48 runs in total. The 108 compound test runs were left unchanged. The same constituent representation, curated feature sets, logistic-regression classifier, crossed outer splits, and grouped inner-CV threshold-calibration procedure were then repeated on the reduced 240-run training-support dataset. Differences from the locked H/L-merged primary analysis were evaluated using 5000 paired physical-run cluster-bootstrap resamples. This experiment was treated only as a sensitivity analysis and was not used to redefine the locked primary method.

E. Classifier-Family Robustness Analysis

To determine whether the observed constituent-generalization patterns were specific to the locked linear diagnostic probe, the complete crossed protocol was repeated with three additional classifier families: a random forest, an RBF-kernel support-vector machine, and a three-hidden-layer feed-forward neural network (MLP). These analyses were performed as post-hoc robustness experiments and were not used to replace or select the locked logistic-regression primary model.

The random forest used 100 trees, square-root feature subsampling, a minimum leaf size of two, and balanced-subsample class weighting. The RBF-SVM used $C = 1$, gamma set to `scale`, and balanced class weighting. The neural baseline used hidden-layer sizes of 64, 32, and 16 units with ReLU activation, Adam optimization, L_2 regularization of 10^{-4}, an initial learning rate of 10^{-3}, and a maximum of 400 iterations. Class balancing for the neural baseline was implemented using balanced sample weights when supported and deterministic balanced oversampling otherwise.

Across classifier families, the constituent labels, curated mechanism-specific feature sets, crossed outer splits, training-only median imputation and standardization, grouped inner cross-validation, threshold search grid, threshold-selection objective, and evaluation metrics were held fixed. No alternative classifier or hyperparameter configuration was selected on the basis of outer-test performance.

F. Closed-Set Control Experiment

An auxiliary closed-set experiment was conducted to establish the conventional diagnostic separability of the dataset before imposing compositional generalization constraints. The original diagnostic states were treated as 24 mutually exclusive classes and evaluated using five-fold cross-validation at the experimental-run level. All windows or derived descriptors belonging to the same physical run were assigned to the same fold. Preprocessing parameters were estimated exclusively from the training runs of each fold, and performance was calculated from out-of-fold predictions so that each physical run contributed to the test set exactly once. This experiment was used only as a closed-set reference and was not used for model selection in the constituent-generalization experiments.

G. Crossed Composition–Context–Condition Generalization Protocol

To separate the effects of novel fault combinations, compound-context exposure, and operating-condition shift, a crossed outer evaluation protocol was constructed over all available compound runs. The compound structure can equivalently be represented as an undirected graph

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}), \tag{15}$$

where the vertex set $\mathcal{V} = \mathcal{F}$ contains the elementary fault mechanisms and an edge $(A, B) \in \mathcal{E}$ indicates that constituents A and B occur together in an observed compound composition. The three context regimes can therefore be interpreted as different graph-withholding operations: recombination removes only the target edge (A, B), one-sided context zero-shot removes all training edges incident to one target constituent, and two-sided context zero-shot removes all training edges incident to either target constituent. Figure 1 illustrates these operations on the compound-composition subgraph.

Let the target test composition be

$$\mathcal{C}^* = \{A, B\}. \tag{16}$$

Three levels of compound-context exposure were considered.

1) Recombination: The exact pair $A+B$ was excluded from training, while both A and B remained observable in other compound compositions. If $d_{\mathrm{cmp}}(f)$ denotes the number of distinct compound contexts containing constituent f in the outer training set, the recombination regime satisfies

$$d_{\mathrm{cmp}}(A) > 0, \qquad d_{\mathrm{cmp}}(B) > 0. \tag{17}$$

Thus, the constituent mechanisms were familiar in compound operation, but their particular edge or pairing was novel.

2) One-Sided Compound-Context Zero-Shot: In this regime, one constituent was completely removed from all compound contexts while remaining available as a single fault. Without loss of generality,

$$d_{\mathrm{cmp}}(A) = 0, \qquad d_{\mathrm{cmp}}(B) > 0. \tag{18}$$

Both directional cases, A-unseen and B-unseen, were evaluated separately.

3) Two-Sided Compound-Context Zero-Shot: The most restrictive context regime removed all compound examples containing either constituent,

$$d_{\mathrm{cmp}}(A) = 0, \qquad d_{\mathrm{cmp}}(B) = 0, \tag{19}$$

while preserving their eligible single-fault examples. Consequently, neither constituent had been observed in any compound context before testing on $A+B$.

Each composition-context regime was further crossed with operating-condition exposure. Let c_i denote the operating-condition identifier of run i. For the seen-condition setting,

$$c_{\mathrm{test}} \in \mathcal{C}_{\mathrm{train}}, \tag{20}$$

whereas for the unseen-condition setting,

$$c_{\mathrm{test}} \notin \mathcal{C}_{\mathrm{train}}. \tag{21}$$

The complete experimental design therefore consisted of

$$3 \text{ compound-context regimes} \times 2 \text{ condition regimes}, \tag{22}$$

resulting in six principal evaluation scenarios.

All splitting was performed at the experimental-run level before any classifier fitting. Composition identities were defined from the constituent vector rather than severity-specific class names, thereby preventing severity variants of the same constituent combination from unintentionally leaking across the composition boundary. Because either member of a target compound could be deprived of prior compound-context exposure, the one-sided regime was evaluated in both directions. Consequently, this regime produced 216 prediction rows from 108 unique physical compound runs, whereas the recombination and two-sided regimes each produced 108 prediction rows from 108 unique runs. The two directional predictions associated with the same physical run were never treated as independent experimental units in uncertainty estimation; all bootstrap analyses were clustered by physical run.

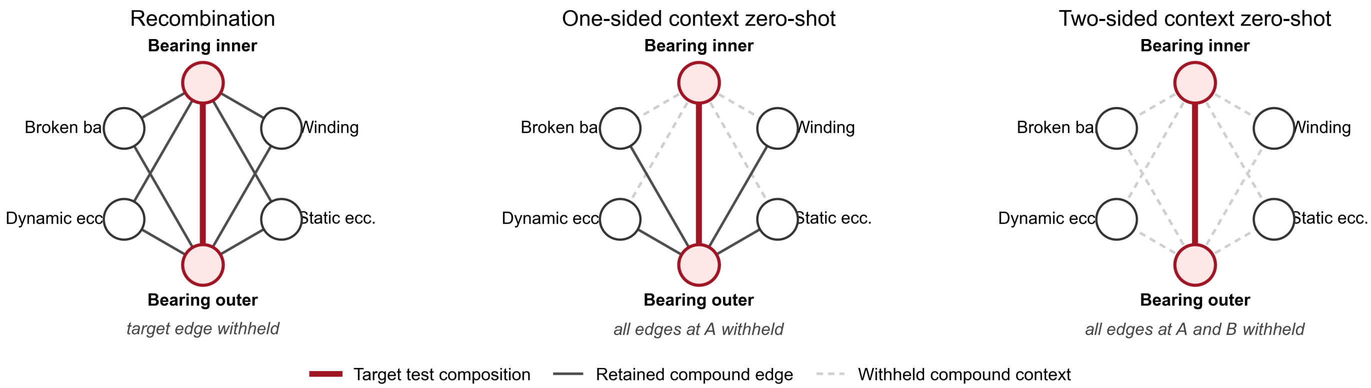

Fig. 1. Graph interpretation of the three compound-context evaluation regimes. Nodes denote fault mechanisms participating in the compound subgraph, edges denote observed compound compositions, and the red edge denotes the target test composition. Recombination withholds only the target edge; one-sided context zero-shot removes all compound-context exposure for one target constituent; and two-sided context zero-shot removes compound exposure for both target constituents.

H. Evaluation Metrics

Performance was evaluated using metrics that directly quantify recovery of the constituent set rather than only conventional sample-level classification. The overall constituent recall was defined as

$$R_{\text{const}} = \frac{\sum_{i=1}^{N} \sum_{k=1}^{K} y_{i,k}\hat{y}_{i,k}}{\sum_{i=1}^{N} \sum_{k=1}^{K} y_{i,k}}. \tag{23}$$

Exact constituent-set recovery was computed as

$$\text{EM} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}\left(\hat{\mathbf{y}}_i = \mathbf{y}_i\right). \tag{24}$$

The average number of incorrectly added mechanisms per prediction row was

$$\text{FA} = \frac{1}{N} \sum_{i=1}^{N} \sum_{k=1}^{K} (1 - y_{i,k})\, \hat{y}_{i,k}. \tag{25}$$

The fraction of compound predictions for which at least one true constituent was recovered was

$$R_{\text{any}} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}\left(\sum_{k=1}^{K} y_{i,k}\hat{y}_{i,k} > 0 \right), \tag{26}$$

whereas recovery of all true constituents, irrespective of additional false activations, was defined as

$$R_{\text{all}} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}\left(\sum_{k=1}^{K} y_{i,k}\hat{y}_{i,k} = \sum_{k=1}^{K} y_{i,k} \right). \tag{27}$$

Micro-averaged F_1 was calculated over all constituent decisions as

$$F_{1,\text{micro}} = \frac{2TP}{2TP + FP + FN}. \tag{28}$$

Because some evaluation subsets contained only a subset of the nine constituents, macro-F_1 was additionally calculated over active constituents,

$$F_{1,\text{active}} = \frac{1}{|\mathcal{K}_+|} \sum_{k \in \mathcal{K}_+} F_{1,k}, \tag{29}$$

where $\mathcal{K}_+$ denotes the constituents having at least one positive test instance.

I. Within-Pair Detectability and Cross-Partner Transfer

To determine whether poor constituent recovery reflected an absence of measurable evidence, an oracle detectability analysis was performed. For constituent A in target compound $A{+}B$, the compound state was compared directly with the corresponding single-B state,

$$A{+}B \quad \text{vs.} \quad B. \tag{30}$$

The resulting within-pair discriminability was quantified using ROC AUC and denoted $\text{AUC}^{\text{within}}_{A|B}$.

A stronger transfer analysis evaluated whether evidence for A learned from eligible compound contexts other than the target composition remained useful when A occurred with partner B. The target pair $A{+}B$ was excluded from training and the resulting fixed-orientation score was evaluated on $A{+}B$ vs. B. The conventional directional cross-partner AUC was denoted $\text{AUC}^{A|B}_{\text{dir}}$.

Because an AUC below 0.5 can reflect reversed ranking rather than absence of separability, an orientation-free diagnostic measure was additionally defined as

$$\text{AUC}_{\text{sep}} = \max\left(\text{AUC}_{\text{dir}}, 1 - \text{AUC}_{\text{dir}}\right) = 0.5 + \left|\text{AUC}_{\text{dir}} - 0.5\right|. \tag{31}$$

AUC_{sep} was used only to characterize rank separability and was not interpreted as deployable classification performance, because post-hoc inversion of an unknown target score would require target-label information. The fixed-orientation transfer gap was defined as

$$\Delta_{\text{transfer}} = \text{AUC}^{\text{within}}_{A|B} - \text{AUC}^{A|B}_{\text{dir}}. \tag{32}$$

This quantity characterizes degradation of the learned score orientation and should not be interpreted directly as loss of orientation-free separability.

Because the released recordings were acquired across multiple recording dates, an additional recording-sensitivity analysis was performed. Acquisition timestamps were recovered from the original file names for 287 of the 288 runs, spanning seven recording dates. The only duplicated fault–condition state available in two distinct recordings was used as a nuisance-control experiment. In addition, compound pairs acquired on the same date and sharing one constituent were compared under matched operating conditions to reduce recording-date mismatch in the interpretation of within-pair separability. These analyses were treated as sensitivity controls rather than as causal estimates of acquisition-session effects.

J. Physical-Signature Preservation Analysis

A complementary feature-level analysis characterized how predefined physical signatures changed when a constituent was embedded in a compound fault. Only the mechanism-specific curated feature families defined independently of the cross-partner results were used. For feature j associated with constituent A, let δ_j^A denote its signed standardized evidence under the single-fault reference and $\delta_j^{A|B}$ denote the evidence for the same feature when A occurred together with partner B. Feature preservation was summarized as

$$\rho_j^{A|B} = \frac{\delta_j^{A|B}}{\delta_j^A}. \tag{33}$$

Negative values indicate feature-level sign reversal, values between zero and one indicate suppression, values around one indicate approximate preservation, and values greater than one indicate amplification. This analysis was used to characterize heterogeneity of predefined physical signatures and was not interpreted as a causal decomposition of the multivariate classifier score.

K. Statistical Analysis

Uncertainty of the principal constituent-level metrics was estimated using nonparametric cluster bootstrap resampling at the physical-run level. In the one-sided regime, both directional predictions associated with a physical run were retained within the same bootstrap cluster. In addition to protocol-specific confidence intervals, paired bootstrap contrasts were computed because the same physical compound runs contributed to the compared context regimes. Three prespecified contrasts were evaluated separately under seen and unseen operating conditions:

$$\Delta_{R-O} = M_{\text{recomb}} - M_{\text{one}}, \tag{34}$$

$$\Delta_{R-T} = M_{\text{recomb}} - M_{\text{two}}, \tag{35}$$

$$\Delta_{O-T} = M_{\text{one}} - M_{\text{two}}, \tag{36}$$

where M denotes the metric of interest. The 95% confidence interval was obtained from the 2.5th and 97.5th percentiles of the paired bootstrap distribution. A contrast was considered resolved when this interval excluded zero. To formally test whether the effect of compound-context deprivation changed under operating-condition shift, a difference-of-differences statistic was also evaluated,

$$\Delta_{\text{int}} = \Delta_{\text{unseen}} - \Delta_{\text{seen}}, \tag{37}$$

where Δ denotes the paired difference between recombination and the corresponding context-zero-shot regime. Confidence intervals were estimated using 5000 paired cluster-bootstrap resamples at the physical-run level. For the auxiliary closed-set experiment, 95% confidence intervals were estimated using

TABLE II
CLOSED-SET MULTICLASS DIAGNOSTIC PERFORMANCE WITH 95% BOOTSTRAP CONFIDENCE INTERVALS.

Metric	Estimate	95% CI
Accuracy	0.9132	$[0.8854, 0.9375]$
Balanced accuracy	0.9134	$[0.8856, 0.9377]$
Macro-F_1	0.9119	$[0.8826, 0.9366]$

5000 stratified run-level bootstrap resamples within the original 24 diagnostic classes.

For cross-partner transfer, evidence for reversed score orientation was tested using

$$T = |\text{AUC}_{\text{dir}} - 0.5|\,. \tag{38}$$

Positive and negative scores were exchanged within matched operating conditions under permutation, thereby preserving the paired experimental structure. Family-wise multiplicity across the 18 directed constituent–partner relations was controlled using the Holm procedure. A relation was described as a statistically supported polarity reversal only when its directional AUC was below 0.5, its bootstrap interval remained below 0.5, and the Holm-adjusted permutation analysis supported departure from chance orientation.

IV. RESULTS

A. Closed-Set Discrimination and the Generalization Gap

Before evaluating compositional generalization, an auxiliary closed-set experiment was conducted to determine whether the dataset was intrinsically difficult to discriminate when all diagnostic classes were represented during training. The 24-class closed-set classifier achieved an accuracy of 0.9132, a balanced accuracy of 0.9134, and a macro-F_1 score of 0.9119, as summarized in Table II.

These results demonstrate strong conventional closed-set discriminability when all diagnostic classes are represented during training. Consequently, the substantially lower constituent-level performance observed under the crossed protocols is unlikely to arise solely from poor baseline separability; rather, it emerges under the additional requirements of composition, compound-context, and operating-condition generalization.

B. Crossed Composition–Context–Condition Generalization

The proposed crossed evaluation protocol was applied to all 108 unique physical compound-fault runs. Each target compound was evaluated under recombination, one-sided compound-context zero-shot, and two-sided compound-context zero-shot regimes, with each regime further separated into seen- and unseen-operating-condition settings. In the one-sided regime, both directional removals were evaluated because either constituent could be the mechanism deprived of compound-context exposure. Table III summarizes the principal constituent-level results.

When both constituents retained prior exposure to other compound contexts, previously unseen pairings remained partially recoverable. Under the recombination protocol, micro-F_1 was 0.6775 under seen operating conditions and 0.6865

TABLE III
CONSTITUENT-LEVEL GENERALIZATION PERFORMANCE UNDER CROSSED COMPOSITION-CONTEXT AND OPERATING-CONDITION REGIMES. EM DENOTES EXACT CONSTITUENT-SET RECOVERY, R_{const} DENOTES CONSTITUENT RECALL, FA/RUN DENOTES THE MEAN NUMBER OF INCORRECTLY ADDED INACTIVE CONSTITUENTS PER COMPOUND PREDICTION ROW, AND R_{any} AND R_{all} DENOTE RECOVERY OF AT LEAST ONE AND ALL TRUE CONSTITUENTS, RESPECTIVELY.

Condition	Context regime	Active Macro-F_1	Micro-F_1	EM	R_{const}	FA/run	R_{any}	R_{all}
Seen	One-sided	0.5890	0.6581	0.1204	0.6505	0.6528	0.9259	0.3750
Seen	Recombination	0.6177	0.6775	0.1389	0.6759	0.6389	0.9352	0.4167
Seen	Two-sided	0.5994	0.6620	0.1111	0.6620	0.6759	0.9444	0.3796
Unseen	One-sided	0.5761	0.6452	0.1111	0.6481	0.7222	0.9398	0.3565
Unseen	Recombination	0.6177	0.6865	0.1574	0.6944	0.6574	0.9444	0.4444
Unseen	Two-sided	0.5588	0.6311	0.0833	0.6296	0.7315	0.9352	0.3241

TABLE IV
INPUT-INDEPENDENT CONSTANT-PREDICTOR REFERENCES ON THE 108 COMPOUND RUNS. THESE REFERENCES QUANTIFY PERFORMANCE OBTAINABLE FROM CONSTITUENT PREVALENCE ALONE.

Constant prediction	Micro-F_1	EM	R_{const}	FA/run	R_{any}	R_{all}
Inner + outer	0.5556	0.1111	0.5556	0.8889	1.0000	0.1111
Inner + outer + broken bar	0.5333	0.0000	0.6667	1.6667	1.0000	0.3333
Inner only	0.3704	0.0000	0.2778	0.4444	0.5556	0.0000

when the operating condition was also unseen. Constituent recall showed a similar pattern, reaching 0.6759 and 0.6944, respectively.

Exact constituent-set recovery remained limited, ranging from 0.0833 to 0.1574 across the six crossed regimes. However, the trivial references in Table IV show that some apparently favorable partial-recovery metrics are strongly influenced by constituent prevalence. In particular, the constant inner-plus-outer predictor achieves $R_{\text{any}} = 1.0$ without using the input signals. Consequently, R_{any} is retained only as a descriptive measure and is not interpreted as evidence of diagnostic skill.

Constituent recall alone is not sufficient to establish diagnostic skill in this setting. The constant inner-plus-outer-plus-broken-bar predictor reaches $R_{\text{const}} = 0.6667$ without using the input signals, but does so at the cost of 1.6667 false additions per run and zero exact matches. Accordingly, constituent recall is interpreted jointly with false additions, micro-F_1, and exact-set recovery throughout the remainder of the analysis.

The locked logistic-regression detector achieved micro-F_1 values of 0.631–0.686 across the six crossed regimes, exceeding the best constant-predictor micro-F_1 of 0.556. At the same time, FA/run remained between 0.639 and 0.732, substantially below the 1.667 false additions produced by the three-constituent constant reference and also below the 0.889 value of the constant inner-plus-outer predictor. Under recombination, R_{all} reached 0.4167 and 0.4444 for seen and unseen conditions, respectively, compared with 0.1111 for the two-constituent constant reference. These results indicate nontrivial constituent discrimination beyond prevalence-driven constant prediction, while the low exact-match rates show that reliable exact decomposition of unseen compound faults remains unresolved.

C. Paired Effects of Compound-Context Deprivation

Paired run-cluster bootstrap analysis was used to determine which differences between the context regimes were supported beyond variation across physical runs. Figure 2 summarizes the prespecified contrasts for constituent recall and micro-F_1.

Under seen operating conditions, the differences between recombination and the context-zero-shot regimes were limited. For constituent recall, the recombination–one-sided contrast was 0.0255 with a 95% CI of $[0.0000, 0.0509]$, whereas the recombination–two-sided contrast was 0.0139 with a 95% CI of $[-0.0186, 0.0509]$. The corresponding micro-F_1 contrasts also included zero.

A different pattern was observed when the target operating condition was withheld. Relative to one-sided context zero-shot, recombination improved constituent recall by 0.0463 (95% CI: $[0.0231, 0.0694]$) and micro-F_1 by 0.0413 (95% CI: $[0.0223, 0.0636]$). Relative to two-sided context zero-shot, the recombination advantage was 0.0648 for constituent recall (95% CI: $[0.0278, 0.1065]$) and 0.0554 for micro-F_1 (95% CI: $[0.0232, 0.0895]$).

The one-sided–two-sided differences remained unresolved for both metrics under unseen operating conditions. A formal difference-of-differences analysis further showed that the penalty associated with complete compound-context deprivation increased under operating-condition shift. For recombination versus two-sided context zero-shot, the interaction difference was 0.0400 for micro-F_1 (95% CI: $[0.0088, 0.0731]$), 0.0509 for constituent recall (95% CI: $[0.0185, 0.0880]$), and 0.0833 for all-constituents recovery (95% CI: $[0.0278, 0.1481]$). In contrast, the recombination-versus-one-sided interaction was not resolved for micro-F_1 or constituent recall, although the interaction for all-constituents recovery was positive at 0.0463 (95% CI: $[0.0046, 0.1019]$). These findings indicate that operating-condition shift amplifies the penalty of complete compound-context deprivation rather than producing a uniform interaction across all context

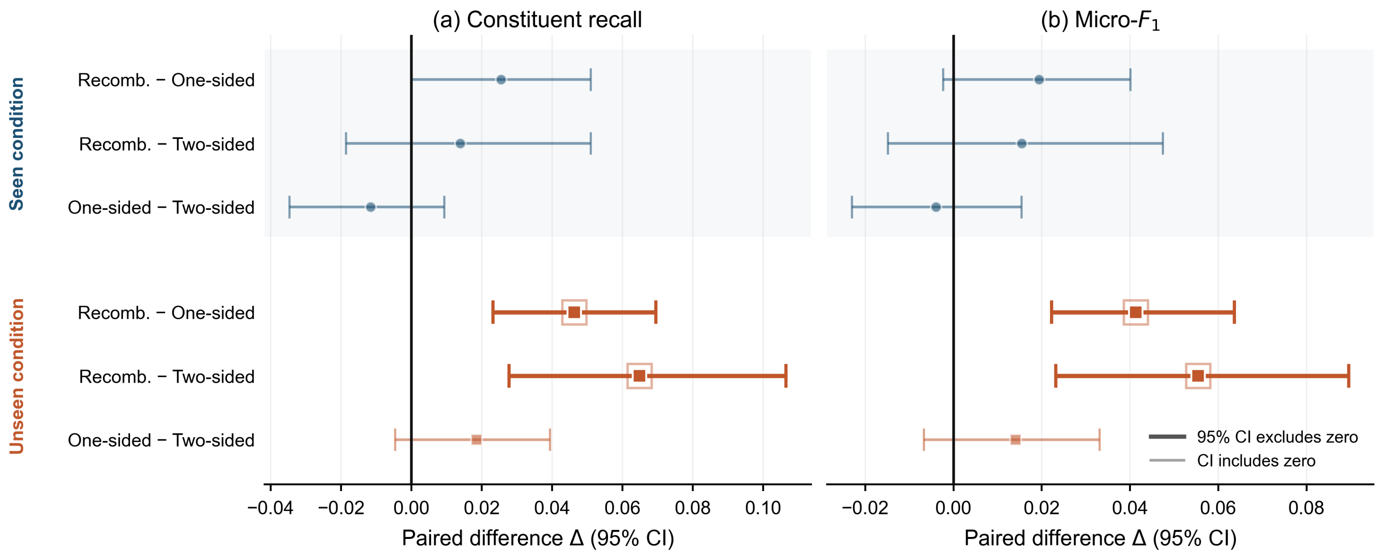

Fig. 2. Paired run-cluster bootstrap differences between composition-context regimes. Points denote paired differences in constituent recall and micro-F_1, and horizontal bars denote 95% bootstrap confidence intervals. Positive values favor the first regime in each contrast.

regimes.

Directional decomposition of the one-sided regime revealed substantial mechanism dependence. When broken-bar compound context was withheld, constituent recall remained high at 0.979 under seen conditions and 0.896 under unseen conditions. In contrast, withholding winding compound context reduced recall to 0.438 and 0.479, respectively, while withholding dynamic-eccentricity context yielded a recall of 0.500 under both condition settings. Each of the 108 physical compound runs contributed exactly two directional one-sided predictions, and all bootstrap uncertainty estimates remained clustered by physical run.

D. Sensitivity to Decision-Threshold Objective

The limited exact-match performance and non-negligible false-addition counts motivated a post-hoc sensitivity analysis of the inner-CV threshold objective. The original F_1-based thresholds were retained as the locked primary analysis. Alternative $F_{0.5}$ and balanced-accuracy objectives were evaluated only to characterize the precision–recall operating-point tradeoff and were not used for post-hoc model selection.

Precision-weighted $F_{0.5}$ calibration substantially reduced false additions. Across the six crossed regimes, FA/run decreased from 0.639–0.731 under the locked F_1 thresholds to approximately 0.222–0.389, while exact-match recovery increased from 0.083–0.157 to 0.157–0.287. This gain was accompanied by lower constituent recall, which decreased to approximately 0.583–0.644. Conversely, balanced-accuracy thresholding increased constituent recall but also increased false additions.

These results show that exact compound decomposition is strongly affected by the operating point of the constituent detectors. Precision-oriented thresholding suppresses overdiagnosis and improves exact-set recovery, whereas recall-oriented thresholding recovers more true constituents at the cost of additional constituent activations. As shown by the classifier-family robustness analysis below, absolute decomposition quality is also model-dependent. Consequently, the low exact-match performance of the locked linear probe should not be interpreted as an intrinsic upper bound imposed by the crossed protocol itself.

E. Sensitivity to Single-Fault Severity Merging

Removing low-severity single-fault support did not produce a uniform change in constituent recall, exact-match recovery, or all-constituent recovery across the crossed regimes. Most paired differences for these metrics had 95% confidence intervals spanning zero, indicating that the principal generalization findings were not driven by pooling low- and high-severity single-fault examples.

The clearest differences were observed in false additions. Under seen conditions, removing low-severity support increased FA/run from 0.653 to 0.755 in the one-sided regime ($\Delta = +0.102$, 95% CI: $[0.046, 0.162]$) and from 0.639 to 0.722 under recombination ($\Delta = +0.083$, 95% CI: $[0.019, 0.157]$). Under unseen conditions, the two-sided regime similarly increased from 0.731 to 0.843 FA/run ($\Delta = +0.111$, 95% CI: $[0.037, 0.185]$). For seen-condition recombination, micro-F_1 decreased from 0.677 to 0.651 when low-severity support was removed ($\Delta = -0.026$, 95% CI: $[-0.052, -0.002]$). Other micro-F_1 differences were unresolved. Overall, the sensitivity analysis indicates that severity merging does not account for the principal crossed-generalization pattern and, in several regimes, the additional low-severity support reduced over-diagnosis rather than artificially inflating constituent recovery.

F. Robustness Across Classifier Families

The complete crossed protocol was additionally evaluated with random-forest, RBF-SVM, and neural classifiers while

TABLE V
DESCRIPTIVE CLASSIFIER-FAMILY ROBUSTNESS AVERAGED ACROSS THE SIX CROSSED EVALUATION SCENARIOS. LR DENOTES LOGISTIC REGRESSION AND MLP DENOTES THE THREE-HIDDEN-LAYER FEED-FORWARD NEURAL BASELINE.

Model	Micro-F_1	EM	R_{const}	FA/run
LR	0.660	0.120	0.660	0.680
RF	0.744	0.282	0.669	0.256
RBF-SVM	0.692	0.157	0.664	0.509
MLP	0.737	0.310	0.659	0.256

retaining the same constituent representation, curated feature families, outer evaluation splits, and inner-CV threshold-calibration principles. Table V summarizes descriptive performance averaged across the six crossed scenarios. The locked logistic-regression results remain the primary analysis; the alternative classifiers are reported only as robustness references.

Absolute decomposition performance was classifier dependent. Random forest achieved the highest mean micro-F_1 (0.744), whereas the neural baseline achieved the highest mean exact-match rate (0.310). Both random forest and the neural baseline reduced mean FA/run to 0.256, compared with 0.680 for the locked logistic-regression probe. In contrast, mean constituent recall varied only from 0.659 to 0.669 across the four classifiers. Thus, classifier family primarily affected the precision of constituent-set formation and the number of additional activations rather than the overall fraction of true constituents recovered.

The qualitative recombination advantage was nevertheless retained across all four classifier families. Under unseen operating conditions, the recombination–two-sided difference in micro-F_1 was 0.055 for logistic regression, 0.051 for random forest, 0.032 for RBF-SVM, and 0.023 for the neural baseline. The corresponding constituent-recall differences were 0.065, 0.065, 0.069, and 0.032, respectively. These descriptive results indicate that the adverse effect of complete compound-context deprivation is not unique to the locked linear classifier.

G. Oracle Detectability and Cross-Partner Constituent Transfer

To determine whether unsuccessful constituent recovery resulted from the physical disappearance of a fault signature, a within-pair oracle analysis was first performed. For each directed compound relation $A|B$, measurements from the compound state $A{+}B$ were compared with the corresponding single-B reference while retaining the same constituent of interest A.

All 18 directed constituent–partner tests exhibited strong within-pair discriminability. The condition-held-out within-pair AUC ranged from 0.9167 to 1.000, with a median of 1.000. Thus, the target compound and its corresponding partner-only reference remained strongly separable in the broader physics-feature space. However, the recording-sensitivity control showed that the two available recordings of the same bearing-outer fault under the same operating condition were themselves nearly perfectly separable ($\mathrm{AUC} = 0.998$). High within-pair AUC therefore cannot be attributed uniquely to the additional constituent and may partly reflect recording-specific nuisance variation.

A complementary same-date control reduced recording-date mismatch. Eight pairs of compound states acquired on the same date shared one constituent and all 12 operating conditions, yielding 16 directed matched comparisons. Leave-one-condition-out AUCs ranged from 0.972 to 1.000. In particular, static eccentricity in the bearing-outer context remained perfectly separable when the alternative same-date compound contained either broken bar or winding with the same bearing-outer partner ($\mathrm{AUC} = 1.000$ in both comparisons). Because these reference compounds contain a different non-target constituent, the controls do not causally isolate the target mechanism; nevertheless, they show that the observed compound-state discriminability cannot be explained solely by recording-date mismatch.

Accordingly, within-pair detectability does not establish that the same evidence is transferable across compound partners. Cross-partner experiments were therefore performed by learning constituent A from other compound contrasts, such as $A{+}C$ versus C, and evaluating the resulting detector on $A{+}B$ versus B.

Under seen operating conditions, the primary cross-partner protocol achieved a median AUC of 0.905 across the 18 directed tests, with a mean AUC of 0.815. Thirteen of the 18 directed tests achieved an AUC of at least 0.80, and 15 achieved an AUC of at least 0.70. When the target operating condition was also withheld, the median AUC remained high at 0.884 and the mean AUC was 0.803.

These aggregate values indicate that constituent evidence is often transferable across partners. Nevertheless, the transfer behavior was highly heterogeneous. Selected examples are reported in Table VI. The complete pattern of directional cross-partner transfer is shown in Figure 3.

The contrast between within-pair and cross-partner performance was particularly pronounced for static eccentricity in the presence of the bearing-outer fault. Within the target pair, static eccentricity remained strongly detectable, with $\mathrm{AUC}^{\text{within}} = 0.9167$. However, evidence learned from other eligible compound contexts produced a fixed-orientation directional AUC of

$$\mathrm{AUC}_{\text{dir}} = 0.000 \tag{39}$$

under both seen and unseen operating conditions. The corresponding orientation-free separability was

$$\mathrm{AUC}_{\text{sep}} = 1.000, \tag{40}$$

indicating perfect rank separability in the opposite orientation. This was the only directed constituent–partner relation classified as a statistically supported score-orientation reversal after multiplicity correction, with Holm-adjusted $p = 0.0136$ under seen conditions and $p = 0.0108$ under unseen conditions. The statistical test establishes a reproducible reversal of the fixed-score ordering, but it does not establish that the reversal is caused exclusively by a physical inversion of the static-eccentricity signature. Given the strong recording-identity separability observed in the nuisance control, partner-dependent

TABLE VI
SELECTED DIRECTED CROSS-PARTNER TRANSFER RESULTS. THE WITHIN-PAIR AUC MEASURES DETECTABILITY OF CONSTITUENT A IN THE TARGET PAIR, WHEREAS THE CROSS-PARTNER AUC MEASURES TRANSFER OF EVIDENCE LEARNED FROM OTHER COMPOUND PARTNERS.

Target composition	Constituent A	Partner B	Within-pair AUC	Cross-partner AUC (seen)	Cross-partner AUC (unseen)
Bearing inner + bearing outer	Bearing outer	Bearing inner	1.000	1.000	1.000
Bearing inner + dynamic eccentricity	Bearing inner	Dynamic eccentricity	1.000	0.944	0.931
Bearing inner + broken bar	Broken bar	Bearing inner	1.000	0.893	0.851
Bearing inner + dynamic eccentricity	Dynamic eccentricity	Bearing inner	1.000	0.711	0.653
Bearing outer + static eccentricity	Static eccentricity	Bearing outer	0.917	0.000	0.000
Bearing outer + winding	Winding	Bearing outer	0.917	0.333	0.340

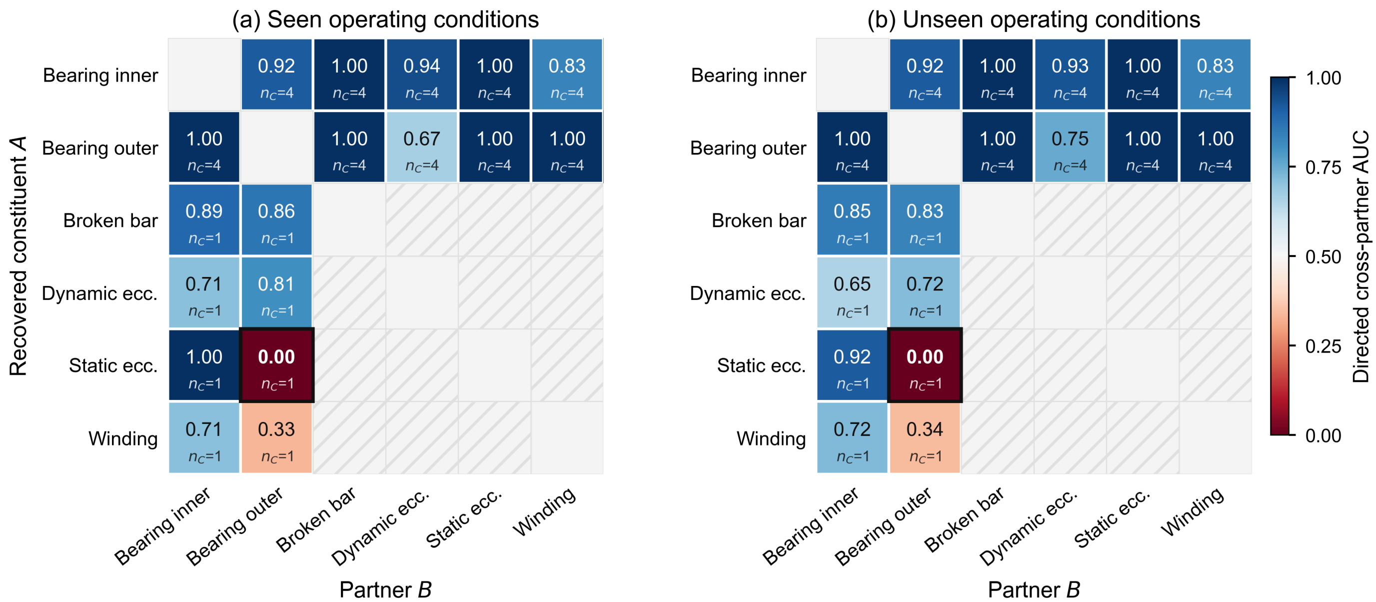

Fig. 3. Directional cross-partner transfer AUC for the primary other-partners transfer protocol. Rows denote the constituent A to be recovered and columns denote partner B in the target compound. Values below 0.5 indicate below-chance fixed-orientation ranking and do not by themselves establish statistically supported polarity reversal. n_C denotes the number of eligible compound training contexts available for the recovered constituent.

constituent reorientation and acquisition-related variation remain plausible, non-exclusive contributors to this result.

Winding in the bearing-outer composition also produced below-chance directional AUCs of 0.3333 and 0.3403 under seen and unseen operating conditions, respectively. However, its orientation-free separability was only approximately 0.67 and the Holm-adjusted polarity tests were not significant. This relation was therefore classified as orientation-ambiguous rather than as a statistically supported score-orientation reversal.

Taken together, the within-pair and recording-sensitivity controls show that compound states can remain strongly discriminable while the orientation of a score learned from other compound partners may fail to transfer. Because recording-specific structure is also detectable in the feature space, this phenomenon is interpreted as context-dependent score transfer rather than as causal proof of physical constituent-signature inversion. Figure 4 directly contrasts within-pair detectability with directional cross-partner transfer across the evaluated constituent–partner relations.

H. Physical-Signature Heterogeneity and Directional Failure Modes

Cross-partner transfer was strongly directional. For example, in the bearing-outer plus static-eccentricity compound, evidence for bearing outer transferred with an AUC of 1.000, whereas evidence for static eccentricity in the opposite direction yielded a directional AUC of 0.000. Transferability was therefore not a symmetric property of the compound pair.

The feature-level preservation analysis further showed that multivariate score-orientation reversal should not be equated with universal inversion of the underlying physical descriptors. For static eccentricity in the bearing-outer context, the median preservation ratio was 0.760, while approximately 18.5% of the eligible curated feature-condition observations exhibited feature-level sign reversal and 31.4% exhibited severe suppression. Thus, most individual physical descriptors did not reverse orientation even though the resulting multivariate cross-partner ranking was completely inverted.

Conversely, strong feature suppression did not necessarily imply score-orientation reversal. Bearing-outer evidence in the bearing-inner compound showed a median preservation ratio of approximately 0.305 and a severe-suppression

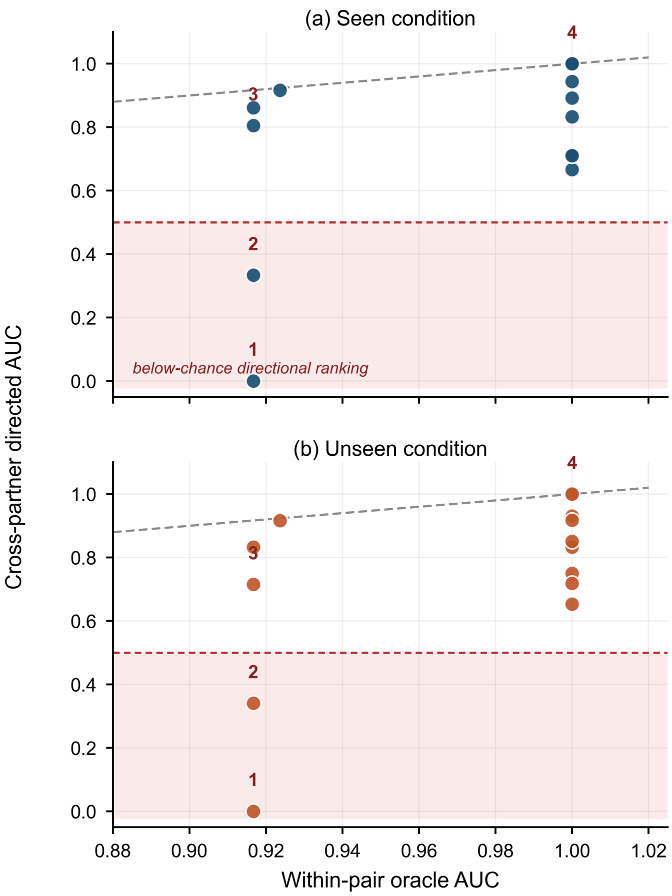

Fig. 4. Within-pair oracle detectability versus directional cross-partner transfer. The dashed horizontal line denotes chance-level directional ranking. The shaded lower region represents below-chance directional ranking; statistically supported polarity reversal requires the separate polarity analysis described in Section III-K.

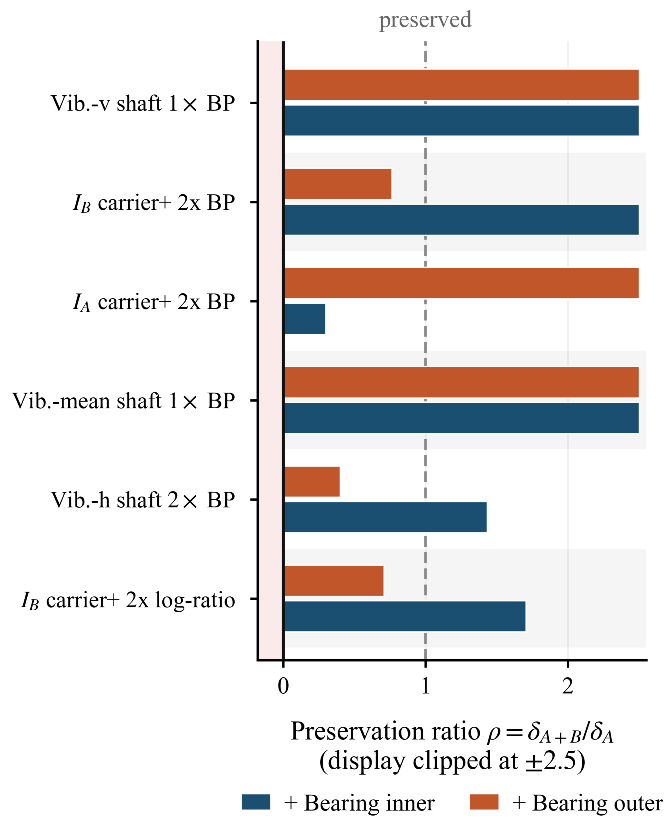

Fig. 5. Feature-level preservation of selected static-eccentricity signatures under bearing-inner and bearing-outer partner contexts. Negative values indicate sign reversal, values near one preservation, and values above one amplification.

rate of approximately 58%, yet its cross-partner directional AUC remained 1.000. Feature-level suppression and multivariate score-orientation reversal therefore represent distinct forms of partner-dependent behavior. Figure 5 illustrates this feature-level heterogeneity for static eccentricity under the two bearing-partner contexts.

For the locked logistic-regression probe, the false-addition values in Table III should not be interpreted as conventional false-alarm probabilities. The metric represents the mean number of inactive constituent labels incorrectly activated per compound prediction. Nevertheless, values of approximately 0.64–0.73 indicate a non-negligible tendency toward over-diagnosis. In a practical condition monitoring system, these additional constituent activations could increase inspection burden even when one or both true mechanisms are successfully identified. Deployment-oriented extensions should therefore consider cost-sensitive threshold calibration or asymmetric decision penalties that explicitly trade constituent recall against false additions. The classifier-family robustness analysis further showed that this over-diagnosis tendency is not fixed by the constituent representation itself: random forest and the neural baseline reduced mean FA/run to approximately 0.26 while retaining mean constituent recall near 0.66.

The observed cross-partner score reorientation also suggests a potential role for representation or normalization strategies that explicitly promote constituent consistency across compound contexts. Possible directions include robust condition-aware normalization learned exclusively from training data, partner-insensitive spectral reference schemes, or representation constraints that align evidence for the same constituent across different compound partners. However, the present preservation analysis shows that the observed failure cannot be reduced to a uniform inversion of the underlying physical features. Such adaptation strategies should therefore be regarded as hypotheses for future investigation rather than as established remedies for score-orientation reversal.

A further limitation is the recording structure of the public dataset. Acquisition timestamps recovered from the released file names showed that distinct recordings of the same fault state can be strongly separable in the physics-feature space. Although same-date shared-partner controls retained near-perfect discriminability, recording date is only a coarse proxy for experimental session, and the available release does not provide replicated same-state measurements sufficient to separate fault, assembly, sensor-mounting, thermal, and recording-session effects. The cross-partner results should therefore be interpreted as evidence of context-dependent score transfer rather than as causal proof of physical signature inversion. Absolute constituent-set reconstruction was additionally sensitive to the classifier family: nonlinear tree and neural baselines substantially reduced false additions and improved exact-match recovery, while the descriptive advantage of recombination over complete compound-context deprivation persisted across all four classifier families.

Taken together, the experiments identify two distinct limitations of constituent-level compound-fault generalization. First, reduced compound-context exposure became measurably detrimental when evaluation also required generalization to an unseen operating condition. Second, cross-partner transfer could be strongly directional even when within-pair constituent evidence remained highly detectable. The latter included one statistically supported case of complete score-orientation reversal, while the available recording structure prevents attributing that reversal uniquely to a physical signature inversion.

V. Conclusion

This paper evaluated whether the elementary mechanisms of a compound motor fault can be recovered when their exact combination, their compound context, and the operating condition are varied independently of the training history. On a public multimodal induction-motor dataset whose compound faults span mechanical and electrical modalities, a locked linear constituent probe over mechanism-specific physical features showed that strong closed-set discriminability (24-class accuracy 0.913, 95% CI $[0.885, 0.938]$) coexists with exact constituent-set recovery of only 0.083–0.157 on unseen compounds. The two results are not in tension; they measure different abilities, and the gap between them is the paper's subject.

Four conclusions follow from the controlled analyses. First, the failure mode is incomplete decomposition rather than loss of information—but this claim survives only when each metric is read against the *best* input-independent constant predictor. At least one true constituent was recovered in more than 92% of compound runs, yet a constant predictor reaches 100% on that metric, so it carries no evidential weight on its own. Against the per-metric floors of Table IV, the locked probe exceeds the recall floor of 0.667 only under recombination (0.676–0.694), exceeds the all-constituent floor of 0.333 under recombination (0.417–0.444) but not in every regime, and does not clearly exceed the exact-match floor of 0.111. Its genuine advantage is the joint operating point—micro-F_1 of 0.631–0.686 against a best-constant 0.556, at comparable or lower false-addition cost—a distinction visible only because every metric was reported against its constant-predictor floor. Second, compound-context deprivation is measurably harmful chiefly when the operating condition is simultaneously unseen: the recombination-versus-two-sided penalty under condition shift was $+0.040$ micro-F_1 (95% CI $[0.009, 0.073]$), $+0.051$ constituent recall, and $+0.083$ all-constituent recovery, whereas the corresponding seen-condition contrasts were not resolved. Third, constituent evidence transfers across compound partners often but directionally: within-pair detectability was near-perfect throughout, yet cross-partner transfer ranged from perfect to a statistically supported score-orientation reversal (Holm-adjusted $p \leq 0.014$)—and the demonstrated separability of distinct recordings of the same fault state requires that reversal to be interpreted as context-dependent score transfer rather than as proven physical signature inversion. Fourth, absolute decomposition quality is an operating-point and model property: precision-weighted thresholding raised exact-match recovery to 0.157–0.287 at reduced recall, and nonlinear classifier families reduced false additions from $\sim$0.68 to $\sim$0.26 per run while the qualitative generalization pattern persisted across all four families.

For practice, these results caution against reading closed-set accuracy as deployment readiness, argue for calibrating constituent detectors with explicit precision–recall cost structure, and indicate that commissioning data should sample compound contexts and operating conditions jointly rather than relying on either alone. The reference experiments of Appendix A reinforce the latter point: on this dataset the standard leave-one-condition-out test is nearly saturated, while single-condition training collapses accuracy, so robustness claims should be evaluated in the extrapolating direction.

Several directions remain open. The dataset provides no replicated same-state recordings that would separate fault, assembly, sensor-mounting, and session effects, so the transfer analyses stop at carefully bounded attribution; replicated acquisitions would convert those bounds into causal statements. Representation or normalization strategies that promote constituent-score consistency across compound partners are identified as hypotheses, not established remedies. Finally, the constituent-level protocol, prevalence references, and controls used here are directly portable to the companion gearbox release and to other multimodal datasets, where cross-dataset replication of the context-by-condition interaction would test its generality.

Appendix A
Operating-Condition Reference Experiments

Section II states that the leave-one-condition-out protocol, the standard robustness test in the multi-condition literature, is nearly saturated on this dataset, whereas the reverse direction is not. This appendix reports the experiments behind that statement. They use a deliberately generic pipeline—distinct from, and simpler than, the constituent-level methodology of Section III—so that the measured gaps reflect protocol difficulty rather than modeling choices, and they are included with the released code.

Windows of 0.64 s (8192 samples, non-overlapping) were described by 108 time- and frequency-domain features (eight temporal statistics and ten spectral descriptors per channel over the three vibration and three current channels), classified into the original 24 diagnostic states by a 300-tree random forest, with three seeds per configuration. Six split protocols were compared. *Random windows* splits windows irrespective of recording (a deliberately leakage-prone reference); *in-condition* trains on the first 60% of every recording and tests on the final 30% with a 10% guard gap; *unknown condition* holds out each of the twelve operating conditions in turn; *cross-profile* trains on one excitation profile (constant-torque variable-speed or constant-speed variable-torque) and tests on

TABLE VII
REFERENCE PROTOCOLS, 24-CLASS ACCURACY (MEAN ± STD OVER FOLDS AND THREE SEEDS). THE SAME FEATURES AND CLASSIFIER ARE USED THROUGHOUT; ONLY THE SPLIT CHANGES.

Protocol	Folds	Accuracy
Random windows (leakage-prone reference)	1	0.975 ± 0.000
In-condition (guard-gap temporal split)	1	0.965 ± 0.000
Unknown condition (leave-one-out)	12	0.939 ± 0.024
Cross-profile	2	0.839 ± 0.115
Steady $\rightarrow$ transitional segments	1	0.766 ± 0.001
Single source (train on one condition)	12	0.333 ± 0.104

the other; *steady → transitional* trains on quasi-stationary segments and tests on speed/load ramps, with stationarity labeled from the key-phase-derived speed and the torque channel (the torque channel is used only for this protocol construction and never as a classifier input); and *single source* inverts the unknown-condition protocol, training on one condition and testing on the remaining eleven.

Table VII summarizes the outcome. Holding one condition out of twelve costs only $\sim$2.6 accuracy points relative to the in-condition reference, because the eleven retained conditions bracket the held-out one in speed and load and the model interpolates. Training on a single condition and extrapolating to the other eleven costs over 60 points; paired over the twelve condition folds, the difference between the two directions is 0.606 (paired t-test $p = 7.7 \times 10^{-10}$; Wilcoxon $p = 4.9 \times 10^{-4}$). Transfer between excitation profiles is likewise asymmetric at equal training size: training under variable speed generalizes to variable load (accuracy 0.944), whereas the reverse loses $\sim$21 points (0.734; paired over seeds, $p = 3.3 \times 10^{-5}$). Finally, appending 36 speed-invariant envelope-order descriptors to the feature set leaves the interpolating protocols unchanged ($+0.001$ on unknown-condition, paired $p = 0.84$) but significantly improves the extrapolating one ($+0.047$ on single source, paired $p = 8.4 \times 10^{-4}$), consistent with their construction.

The same generic pipeline was used for a modality-dissociation experiment that motivates the mechanism-specific selectors of Section III-B. Multi-label one-vs-rest random-forest detectors over severity-split components were trained on the single-fault windows and applied to all compound windows using three feature views: vibration-only, current-only, and the concatenation of both. The dissociation was complete: vibration features never detected an electrically expressed constituent (detection rate 0.000 across all eight cross-modal compounds) and current features never detected a mechanical one. Critically, concatenation destroyed rather than combined the two abilities—the winding constituent was detected on 39.3% of compound windows from current features alone and on 0.2% once vibration features were concatenated, the mechanical signature dominating the shared representation. This is the cross-modal suppression referred to in Section II.

Note that the 24-class figures in this appendix come from the generic window-level pipeline described above and are not comparable to the run-level closed-set control of Section IV-A; the proximity of the appendix's leave-one-condition-out accuracy (0.939) to the main closed-set figure (0.913) is coincidental, as the two experiments differ in features, classifier, evaluation unit, and split.

Two implications carry into the main study. Methodologically, robustness to operating conditions should be claimed in the extrapolating direction, which motivates crossing the composition and context regimes with unseen-condition evaluation rather than relying on leave-one-condition-out alone. Practically, excitation diversity during commissioning matters more than data volume in these comparisons—although the single-source contrast also reduces training-set size, so only the size-matched cross-profile asymmetry isolates diversity itself: a training condition in which speed varies transfers to load variation, but not conversely.

DATA AVAILABILITY

The dataset analyzed in this study is publicly available from Mendeley Data at https://doi.org/10.17632/6s3dggj9mw.1 (CC BY 4.0) [34], with its accompanying descriptor article [1]. Acquisition-timestamp metadata used for the recording-sensitivity controls were recovered from the released file names and are included with the analysis code.

CODE AVAILABILITY

The complete analysis code—including the constituent-level pipeline, the crossed evaluation protocol, the constant-predictor references, the recording-sensitivity controls, and the reference experiments of Appendix A—is available from the corresponding author upon reasonable request.

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